

EDU-01 — Student Learning Segmentation

EdTech | Technical Project Brief

Field	EdTech
Client	Online Learning Platform
Track	Field-Based Scenario
Purpose	Design and implement a machine learning solution that groups students with similar learning behaviors to support personalized educational interventions.

1. Client Background

An online education platform serves students with diverse learning behaviors and engagement levels. While some students actively participate in learning activities and consistently access course materials, others demonstrate low engagement or irregular participation. Since all students currently receive similar support, the platform aims to better understand different learning patterns through data-driven student segmentation.

2. Business Problem

The platform lacks a systematic method to identify meaningful learner groups based on behavioral data. Without understanding these different student profiles, educators cannot provide personalized learning support, targeted interventions, or improve student engagement effectively. As a result, communication and educational strategies remain generic instead of addressing students' individual learning behaviors.

3. Requested Solution

Develop an unsupervised machine learning solution that segments students into meaningful groups according to their learning behavior using the xAPI-Edu-Data dataset. The solution should generate interpretable learner profiles that help educators and product teams design personalized interventions, improve student engagement, and support better educational outcomes.

4. Available Information

The selected xAPI-Edu-Data dataset provides multiple behavioral and demographic variables describing students' learning activities in an online learning environment.

The available information includes:

- Student participation (RaisedHands)
- Learning resource usage (VisitedResources)
- Announcement viewing activity (AnnouncementsView)
- Discussion participation (Discussion)
- Student attendance (StudentAbsenceDays)
- Parent survey participation (ParentAnsweringSurvey)
- Parent satisfaction with school (ParentschoolSatisfaction)
- Student demographics (Gender, Nationality, StageID, GradeID, SectionID, Topic, Semester, Relation)

The dataset also includes a **Class** variable representing academic performance (High, Middle, Low). This variable will **not** be used during clustering because the objective is to discover natural student groups based solely on learning behavior. Instead, it will be used after clustering to help interpret the discovered learner segments.

This section is one of the **most important** because it shows your technical planning before implementation. Since you're using the **xAPI-Edu-Data** dataset and the goal is **student learning segmentation**, here's a version that is technically sound and suitable for a university capstone.

5. Data & Problem Discovery — Complete Before Modeling

Decision /Question	Student Response
Selected dataset and source	Dataset: xAPI-Edu-Data (Students' Academic Performance Dataset). Source: Kaggle (Original Dataset): https://www.kaggle.com/datasets/aljarah/xAPI-Edu-Data . The dataset is publicly available under the CC BY-SA 4.0 license and contains behavioral, demographic, and engagement-related information collected from an online learning environment.
What does one record / sample represent?	Each record represents one individual student enrolled in an online learning environment. It contains demographic information and learning behavior such as classroom participation, resource usage, attendance, discussion activity, and parental involvement.
Proposed target or ML objective	Machine Learning Objective: Unsupervised Learning (Clustering). The objective is to discover meaningful groups of students with similar learning behaviors without using any predefined segment labels. These clusters will support personalized learning interventions and student success strategies.
Key information available at prediction / inference time	At inference time, the model receives only the behavioral and demographic features of a new student, including RaisedHands, VisitedResources, AnnouncementsView, Discussion, StudentAbsenceDays, StageID, GradeID, SectionID, ParentAnsweringSurvey, ParentschoolSatisfaction, Topic, Semester, Relation, Gender, and Nationality. The model then assigns the student to the most similar discovered cluster. The Class variable is excluded from inference because it represents academic performance rather than learning behavior.
Main data quality issues	The dataset contains no missing values, but several categorical variables require encoding before modeling. Numerical features have different scales and therefore require normalization or standardization. Some features may be correlated (e.g., RaisedHands and Discussion), and demographic variables may introduce bias if not interpreted carefully. The relatively small dataset size (480 students) may also limit the generalizability of the discovered clusters.
Potential leakage risks	The main leakage risk is including the Class variable (High, Middle, Low) during clustering because it represents the student's academic outcome rather than learning behavior. Including it would bias the clustering process. Therefore, the Class column will be excluded from training and used only for post-cluster interpretation and evaluation.
Privacy / fairness / licensing concerns	The dataset is fully anonymized and publicly available for research and educational purposes under the Creative Commons Attribution-ShareAlike (CC BY-SA 4.0) license. No personally identifiable information is included. Demographic variables such as Gender and Nationality may introduce fairness concerns if used to make educational decisions; therefore, the resulting clusters will be interpreted as behavioral profiles rather than labels of student ability or potential.

6. Technical Proposal — Complete Before Full Implementation

Decision / Question	Student Response
ML problem formulation	This project is formulated as an unsupervised machine learning (clustering) problem. The objective is to discover natural groups of students based on their learning behavior without relying on predefined segment labels. The identified clusters will represent different learner profiles that can support personalized educational interventions and product improvements.
Proposed baseline	The baseline model will be K-Means Clustering due to its simplicity, efficiency, and strong performance on numerical datasets after preprocessing. It provides an interpretable starting point for comparing more advanced clustering methods.
Main modeling approach(es) to investigate	The primary model will be K-Means Clustering. Additional clustering algorithms such as Agglomerative Hierarchical Clustering and DBSCAN will be investigated for comparison. Principal Component Analysis (PCA) will be used to reduce dimensionality and visualize the discovered clusters in two dimensions.
Data splitting / validation strategy	Since clustering is an unsupervised learning task, traditional train-test splitting is not required for model training. The dataset will be preprocessed by encoding categorical variables and standardizing numerical features. Model quality will be assessed using internal validation metrics and cluster stability across different random seeds. Once the final clustering model is selected, it will be saved and used to assign new students to existing clusters during inference.
Primary evaluation metric(s) and why	The primary evaluation methods will be the Elbow Method, Silhouette Score, and cluster visualization using PCA. The Elbow Method will help determine the optimal number of clusters by identifying the point where adding more clusters provides diminishing improvements. The Silhouette Score will evaluate how well-separated and cohesive the clusters are, with higher values indicating better clustering quality. PCA visualization will be used to visually inspect whether the discovered clusters are reasonably distinct and interpretable. Together, these methods provide both quantitative and qualitative evaluation of the clustering results.
Expected inference input	The inference input consists of a new student's behavioral and demographic information, including features such as RaisedHands, VisitedResources, AnnouncementsView, Discussion, StudentAbsenceDays, StageID, GradeID, SectionID, Topic, Semester, ParentAnsweringSurvey, ParentschoolSatisfaction, Gender, Nationality, and Relation. The same preprocessing pipeline used during training will be applied before prediction.
Expected inference output	The model will output the assigned cluster ID (e.g., Cluster 0, Cluster 1, Cluster 2, Cluster 3) together with the corresponding learner profile description, such as Highly Engaged Learner, Consistent Learner, Struggling but Active Learner, or At-Risk Learner. The output may also include a short recommendation for educators based on the characteristics of the assigned cluster.

Main technical risks / assumptions

The project assumes that the selected behavioral features accurately represent students' learning behavior. The quality of clustering depends on appropriate feature scaling, encoding, and selection. Different clustering algorithms or random initializations may produce different segmentations, so stability analysis will be performed. Because the dataset contains only 480 students, the discovered clusters may not generalize to every online learning platform. Additionally, the generated clusters represent behavioral patterns rather than definitive measures of student ability or academic potential.

7. Functional Requirements

Requirement	How my project satisfies it
Identify meaningful student groups without relying on a pre-existing segment label.	The project uses K-Means clustering , an unsupervised learning algorithm, to discover natural student groups based on learning behavior. The Class variable is excluded during clustering.
Each segment must be interpretable and described in practical terms.	After clustering, each cluster will be analyzed using feature averages and assigned meaningful names such as Highly Engaged Learners , Consistent Learners , Struggling but Active Learners , and At-Risk Learners .
The segmentation must be supported by quantitative evaluation and qualitative interpretation.	Quantitative evaluation will use the Elbow Method and Silhouette Score , while qualitative interpretation will be based on PCA visualization and analysis of cluster characteristics.
The solution should be able to assign or compare a new student profile to the discovered segmentation approach.	After training, the saved clustering model will receive a new student's behavioral features and assign that student to the most appropriate cluster.
The final output must be useful to a product, mentor, or student-success team.	The discovered learner profiles can help educators provide personalized interventions, identify disengaged students, and improve learning support strategies.

8. Expected Deliverables

Deliverable	How it will be delivered
Working ML solution	A complete Google Colab notebook implementing the full student segmentation pipeline.
Trained model or ML pipeline	A trained K-Means clustering model with preprocessing steps (encoding, scaling, PCA, clustering) saved for inference.
Dataset documentation	Documentation of the xAPI-Edu-Data dataset, source, license, preprocessing decisions, assumptions, and selected features.
Evaluation results	Elbow Method plot, Silhouette Score, PCA cluster visualization, and interpretation of the discovered learner profiles.
Inference interface	A simple prediction function (or optional Streamlit/Gradio interface) that assigns a new student to one of the discovered clusters.
Reproducible repository	A GitHub repository containing the notebook, dataset link, requirements, documentation, and instructions for reproducing the project.
Limitations and recommendations	A section discussing dataset limitations, clustering assumptions, fairness considerations, and suggestions for future improvements.

9. Acceptance Criteria

Acceptance Criterion	How the project satisfies it
The solution can process previously unseen input.	The trained preprocessing pipeline and clustering model can assign a new student to an existing learner segment using the same behavioral features.
The output is meaningful for the client use case.	The model generates interpretable learner profiles that support personalized educational interventions and product decisions.
The methodology and evaluation can be explained and defended.	Every stage of the project—including preprocessing, feature selection, clustering, evaluation, and interpretation—is documented and justified.

The prediction/analysis process is reproducible.	The project includes a reproducible Google Colab notebook, fixed random seeds where appropriate, dependency documentation, and clear execution instructions.
Known limitations, risks, and assumptions are documented.	The final report explains dataset limitations, potential biases, assumptions, and the challenges associated with clustering educational data.
The implementation can be run according to its documentation.	The repository includes a step-by-step README with installation instructions, dataset access, and commands to reproduce all results.

Again, **Sections 10 and 11 are not forms that you fill in**. They are **constraints and design questions** that your mentor expects you to answer through your project and presentation.

Here are strong answers tailored to your chosen **xAPI-Edu-Data** dataset.

10. Constraints

In Scope (What this project includes)

Requirement	How my project satisfies it
Prototype ML solution	A working student segmentation system using K-Means clustering.
Public / legally usable data	The publicly available xAPI-Edu-Data dataset licensed under CC BY-SA 4.0 .
Reproducible inference workflow	A Google Colab notebook with preprocessing, training, model saving, and inference for new students.
Reasonable student-scale deployment	A lightweight clustering model suitable for educational institutions and online learning platforms.

Out of Scope (What this project does NOT include)

Requirement	Project limitation
Direct production integration	The project demonstrates a prototype only and is not integrated into a live Learning Management System (LMS).
Unauthorized private data	Only publicly available, anonymized educational data is used.
Enterprise-scale infrastructure	Large-scale cloud deployment and distributed computing are outside the scope of this capstone.
Claims beyond model validity	The discovered clusters describe learning behavior patterns and should not be interpreted as measures of intelligence, academic ability, or future success.

11. Questions You Must Resolve

1. Which behavioral features actually describe learning behavior rather than final outcomes?

The clustering model will use behavioral features that represent student engagement, including:

- RaisedHands
- VisitedResources
- AnnouncementsView
- Discussion
- StudentAbsenceDays

Additional demographic variables may also be included where appropriate, while the **Class** variable (High, Middle, Low) will be excluded because it represents academic performance rather than learning behavior.

2. How many segments are meaningful, and how will you justify that number?

The optimal number of clusters will be determined using the **Elbow Method** and validated with the **Silhouette Score**.

The final number of clusters will also be selected based on business interpretability, ensuring each learner group has clear educational meaning.

Expected result:

4 learner segments

- Highly Engaged Learners
 - Consistent Learners
 - Struggling but Active Learners
 - At-Risk Learners
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3. How will scaling and highly correlated variables affect the result?

Because K-Means relies on distance calculations, numerical features will be standardized before clustering.

Highly correlated variables will be analyzed using a correlation matrix. If two variables provide nearly identical information, one may be removed to improve clustering quality and reduce redundancy.

4. How will you determine whether a mathematically valid cluster is also useful for the business?

A cluster will be considered useful if:

- students within the cluster share similar learning behaviors,
- the cluster can be clearly interpreted,
- educators can identify practical interventions for that group.

For example, identifying an "At-Risk Learners" cluster allows instructors to provide additional support before students disengage completely.

5. How stable are the segments under different modeling choices or random seeds?

The clustering algorithm will be executed multiple times using different random seeds.

If similar learner groups are consistently produced, the segmentation will be considered stable.

The final model will use a fixed random state to ensure reproducible results.

6. What interventions, if any, could reasonably correspond to each segment?

Cluster	Possible Intervention
Highly Engaged Learners	Offer advanced learning materials and leadership opportunities.
Consistent Learners	Maintain regular learning support and encourage continued engagement.
Struggling but Active Learners	Provide tutoring, personalized feedback, and additional practice resources.
At-Risk Learners	Send early-warning notifications, increase mentor communication, and recommend academic support services.
